

Lecture Scribe Generation Guidelines

L12_S1_A

 PDF

Purpose is to generate exam-oriented lecture notes (lecture scribe) strictly from the provided lecture PPT and my notes. The scribe must serve as a precise reference for exams.

Scope is to use only the content explicitly present in the provided material. Do not add external explanations, background knowledge, interpretations or examples. If a definition, assumption, example, derivation, or proof is not present in the material, do not include it.

Structural constraint :

Follow the exact structure and order of the lecture PPT. Preserve topic sequence and hierarchy.

Example Lecture Scribe (structure only)

Topic:

Sample Topic Title

Definition:

A formal definition as stated in the lecture.

Assumptions:

List of assumptions used.

Key Results:

- Statement of a theorem or result.

Proof:

Step 1: First step as presented.

Step 2: Second step as presented.

Conclusion

Example:

A short worked example exactly as shown in the lecture.

Instruction

Using the same structure as the example above, generate the lecture scribe for the provided lecture PPT and notes. The scribe must:

- Be structured exactly like the example
- Include definitions, assumptions, results, proofs, derivations, and worked examples only if they appear in the provided material
- Use simple, clear language while preserving correctness
- Remain fully aligned with the lecture content and ordering

What a lecture scribe is?

A lecture scribe is a faithful, exam-ready reconstruction of what was taught in class, written so that a student can later rely on it as reference material while studying.

A correct lecture scribe should allow a student to:

- Recall definitions and notation
- Reconstruct proofs and derivations
- Follow example solutions step by step
- Understand logical dependencies between concepts

A lecture scribe should answer the question:
“If I study only this document, can I reliably revise what was taught?”

What it's not.

A lecture scribe is not:

- A creative explanation
- A simplified tutorial
- A textbook rewrite
- A solution manual for unseen problems
- A place to add intuition not discussed in class

Everything included must be traceable to the provided context



Topic:

Transformation of Random Variables

Definition:

The objective is to find the **Probability Density Function (PDF)** or **Cumulative Distribution Function (CDF)** of a new random variable (RV), denoted as Y , which is a function of an existing random variable X ($y = g(x)$). The PDF $f_x(x)$ and CDF $F_x(x)$ of X are known *a priori*.

1. Transformation of Random Variables ($y = g(x)$)

Assumptions:

- $y = g(x)$ is a transformation function.
- The inverse function $x = g^{-1}(y)$ exists.

Key Results:

- **Step-1 (CDF Method):** Find the CDF of Y , $F_Y(y) = P(Y \leq y)$.
- **Step-2 (PDF Derivation):** Differentiate the CDF with respect to y to find $f_y(y)$.

Derivation:

- **Step 1:** $F_y(y) = Pr(Y \leq y) = Pr(g(x) \leq y) = Pr(x \leq g^{-1}(y)) = F_x(g^{-1}(y))$.
 - **Step 2:** $f_y(y) = \frac{d}{dy}[F_x(g^{-1}(y))] = f_x(g^{-1}(y)) \cdot \frac{d}{dy}g^{-1}(y)$.
 - **General Formula:** $f_Y(y) = \frac{f_x(x)}{|\frac{dy}{dx}|}$ at $x = g^{-1}(y)$.
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Example:

Problem: X is a uniformly distributed RV on $(-1, 1)$. Find the PDF of $Y = \sin(\frac{\pi X}{2})$.

Solution:

1. **Given PDF of X :** $f_x(x) = \frac{1}{2}$ for $-1 < x < 1$; else 0.
 2. **Transformation:** $y = \sin(\frac{\pi x}{2}) \Rightarrow x = \frac{2}{\pi} \sin^{-1}(y)$.
 3. **Derivative:** $\frac{dx}{dy} = \frac{2}{\pi} \cdot \frac{1}{1-y^2}$.
 4. **Final PDF:** $f_Y(y) = f_x(x) \cdot |\frac{dx}{dy}| = \frac{1}{2} \cdot \frac{2}{\pi} \cdot \frac{1}{1-y^2} = \frac{1}{\pi} \cdot \frac{1}{1-y^2}$ for $-1 < y < 1$.
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2. Function of Two Random Variables

Topic: Sum of Two Random Variables ($Z = X + Y$)

Assumptions:

- X and Y are independent random variables.

Key Result:

- **Convolution Integral:** $f_z(z) = \int_{-\infty}^{\infty} f_x(x) \cdot f_y(z - x)dx$.

Derivation:

1. **CDF setup:** $F_z(z) = Pr(Z \leq z) = Pr(X + Y \leq z)$.
 2. **Double Integral:** $F_z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{xy}(x, y)dx dy$.
 3. **Leibniz Rule Application:** To find the PDF $f_z(z) = \frac{d}{dz} F_z(z)$, use the Leibniz rule for differentiating under the integral sign.
 - Leibniz Rule: $\frac{d}{dx} \int_{a(x)}^{b(x)} h(x, y)dy = h(x, b(x)) \frac{db}{dx} - h(x, a(x)) \frac{da}{dx} + \int_{a(x)}^{b(x)} \frac{\partial}{\partial x} h(x, y)dy$.
 4. **Result:** $f_z(z) = \int_{-\infty}^{\infty} f_{xy}(z - y, y)dy$. For independent variables, this simplifies to the convolution of f_x and f_y .
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Illustrative Examples for Two RVs:

Example 1: Normal Distribution

- **Given:** $X \sim N(0, 1)$ and $Y \sim N(0, 1)$ are independent.
- **Goal:** Prove $Z = X + Y \sim N(0, 2)$.
- **Setup:** $f_z(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-x^2/2} \cdot e^{-(z-x)^2/2} dx$.
- **Conclusion:** After integration, $Z \sim N(0, \sigma^2)$ where $\sigma^2 = 2$.

Example 2: Exponential Distribution

- **Given:** X and Y are exponential RVs with parameter λ .

- **Setup:** $f_z(z) = \int_{-\infty}^{\infty} \lambda e^{-\lambda x} u(x) \cdot \lambda e^{-\lambda(z-x)} u(z-x) dx.$
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3. Other Transformations

- **Difference:** $Z = X - Y.$
 - $F_z(z) = Pr(X - Y \leq z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z+y} f_{xy}(x, y) dx dy.$
- **Ratio:** $Z = \frac{X}{Y}.$
 - $F_z(z) = Pr(\frac{X}{Y} \leq z).$
- **Magnitude:** $R = \sqrt{X^2 + Y^2}.$

Would you like me to create practice problems based on these transformation formulas?